KmtKids: An Engaging Learning Platform for Children

KmtKids is a dynamic educational platform designed to make learning exciting for children. Through interactive experiences, engaging content, and creative challenges, KmtKids transforms education into an adventure, helping young minds explore, discover, and grow in a fun and engaging way



The Inspiration Behind KMT Kids

User's Perspective

Many children lack access to engaging, quality education. Some learning platforms are paywalled.

Young learners miss interactive experiences, limiting their educational outcomes.

Stakeholders' Perspective

Educators need a structured app to easily release content. A dedicated platform is needed.

Reaching young audiences effectively can be a challenge for educators.

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Why History?

We focus on history because it captures children's interest. History offers fun stories, quizzes, and activities that help kids connect with the past. Exploring history sparks curiosity and encourages learning. It's a great way to create engaging educational content.



Engaging Learning Experience

1 Interactive Stories

Explore the wonders of history through engaging stories that capture your imagination.

2 Interactive Quizzes

Test your knowledge with fun quizzes and see how you rank on a leaderboard.

3 AI-Powered Chatbot

Ask questions about history and get personalized answers that are just right for child age!



compatative advantages

AI-Powered Learning

As a platform designed specifically for children, our generative AI adapts to each child's learning style. We enhance efficiency by tailoring responses based on the learner's input, creating engaging educational experiences that bring history to life and keep kids excited about learning.

Free and Inclusive

By offering KmtKids completely free, we eliminate educational barriers, ensuring that every child, regardless of their background, can explore and enjoy the captivating world of history.

Continuous Growth

We're dedicated to innovation and always looking for ways to expand our educational offerings. As we grow, we'll add new subjects and topics to keep KmtKids a rich learning platform for all children.

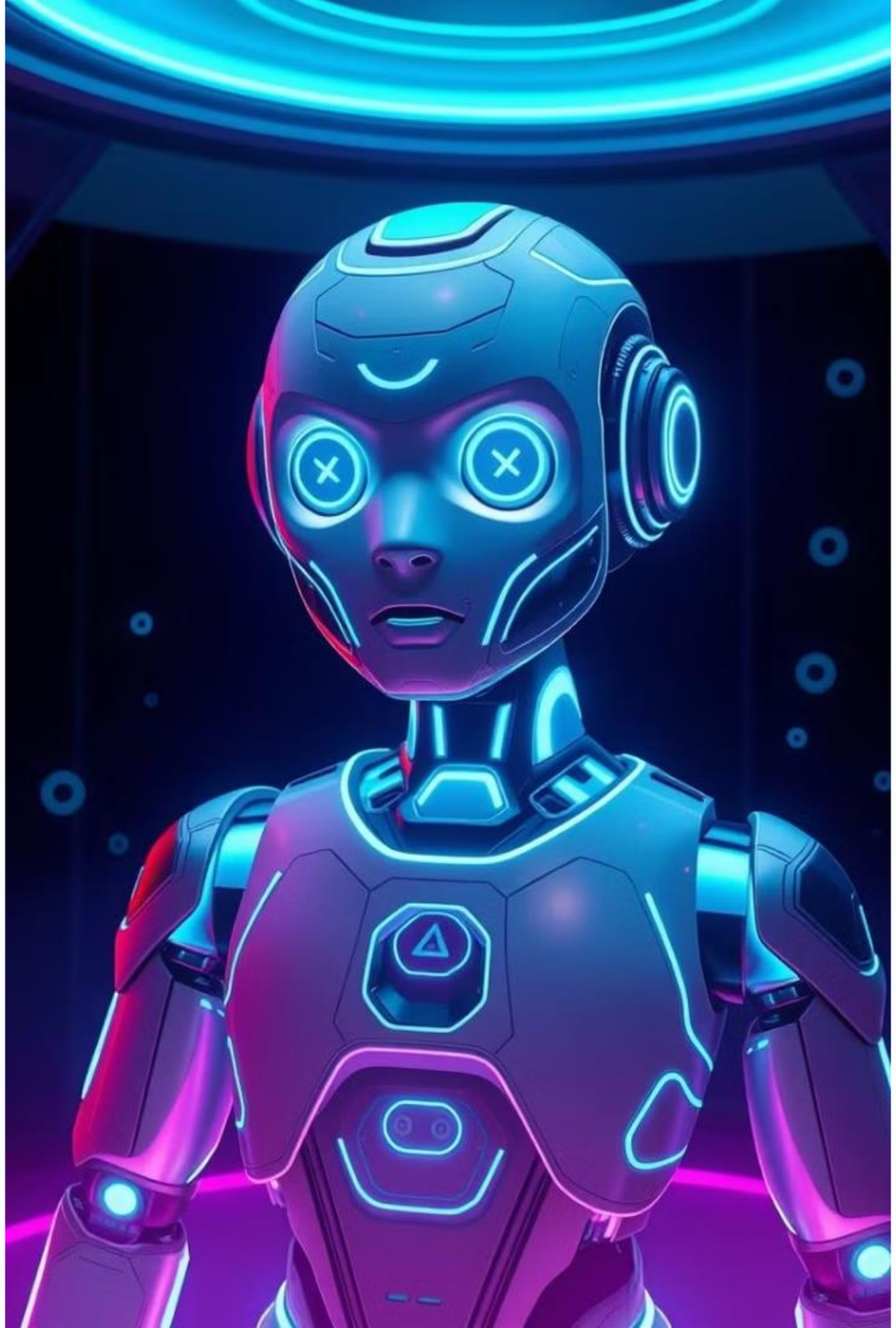
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Future Expansion

mtKids aims to expand beyond history, providing adaptive content that can match various topics

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Gemini Integration: Unleashing the Power of Generative AI

We will explore how to use Gemini, Google's powerful AI, in different apps. You'll see what Gemini can do and how it can improve user experiences on various platforms. With real examples and clear explanations, the goal is to help developers and product managers learn how to use Gemini's AI features effectively.

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Extracting Value from Text

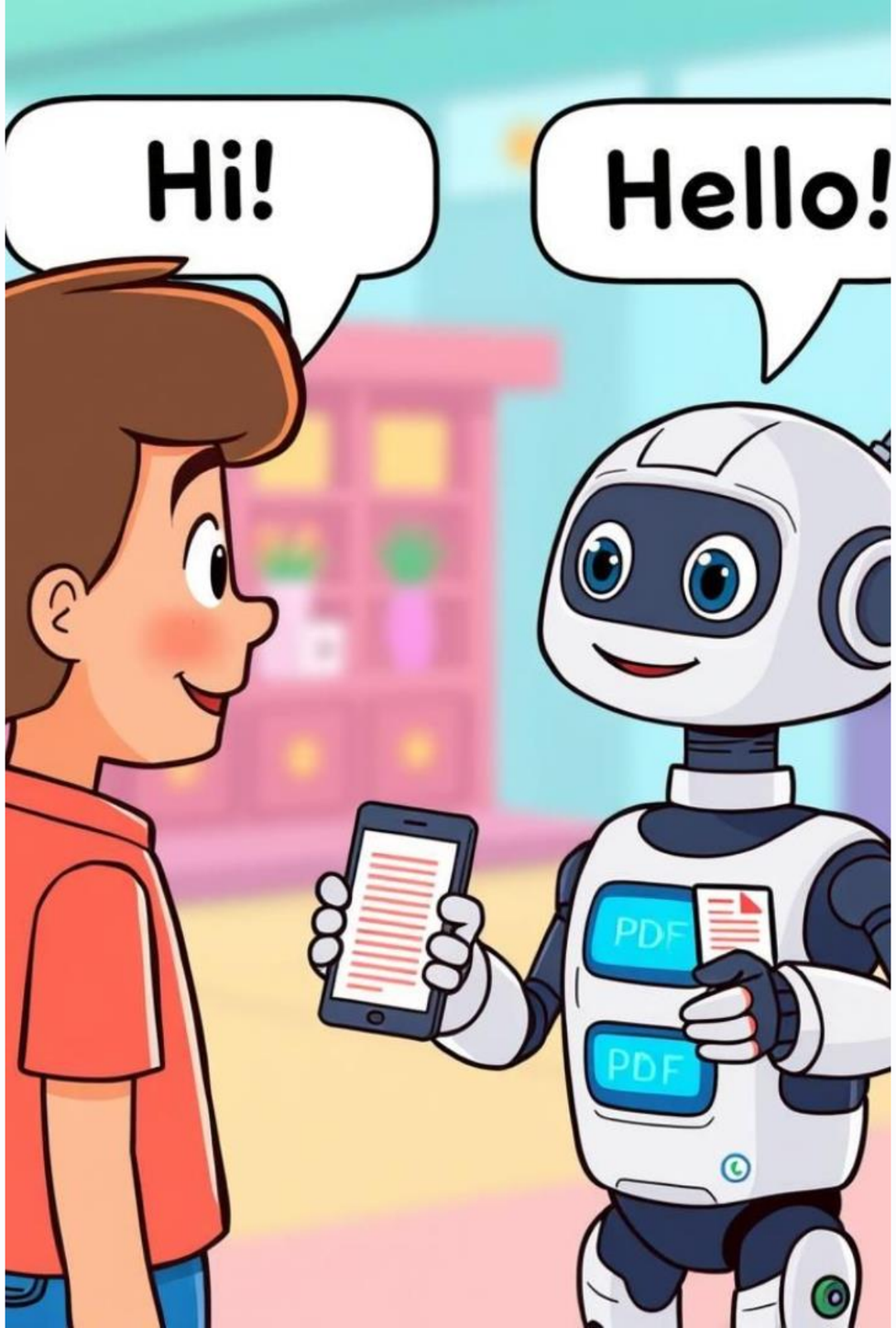
PDF Text Extraction

The system retrieves PDF documents from specified URLs using HTTP requests. It then uses a PDF reading library to extract and concatenate all text from the pages into a unified string, facilitating comprehensive text analysis.

Speech-to-Text Conversion

Using a speech recognition library, the system captures user audio input and converts it into text. This functionality enables processing of Arabic speech, storing the transcribed text for subsequent analysis and application use.

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Powering Conversational AI

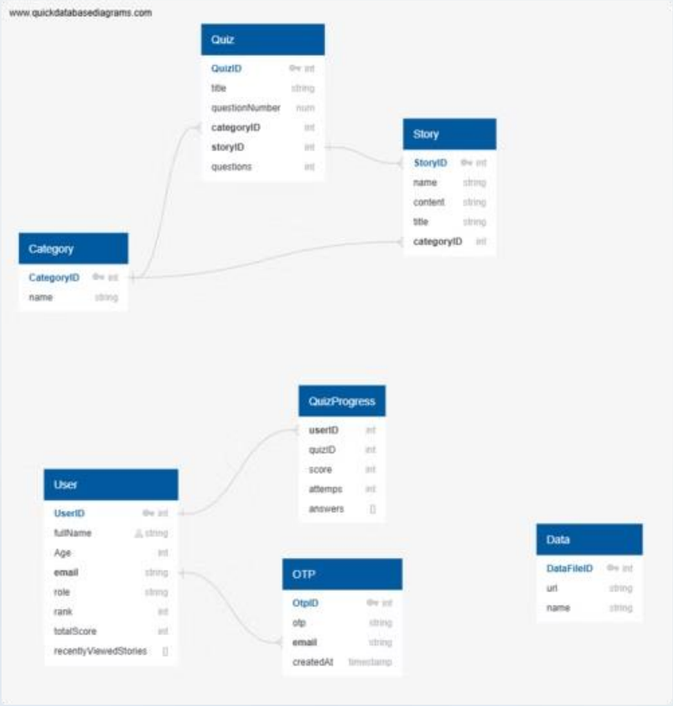
1 Natural Language Processing (NLP)

The Google Generative AI API is used for processing user queries, and the system appends contextual restrictions to guide the AI's response. This ensures that the answers are relevant, informative, and aligned with user needs.

2 AI Chat Session Initialization

A chat session is established to tailor AI responses to the context of extracted PDF content. The extracted text serves as context for the AI model, enabling it to provide responses that are directly relevant to the document.

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Backend Architecture for kmtKids

This part will delve into the backend architecture of our educational platform, highlighting its core functionalities, API endpoints, and deployment infrastructure. We will explore how the backend supports user interactions, content delivery, and data processing, ultimately contributing to a seamless learning experience for users.

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Core Backend Responsibilities

Data Management & Storage

The backend utilizes MongoDB for structured data storage, leveraging Mongoose as the ODM (Object Data Modeling) library. Mongoose provides solutions for maintaining data consistency, including data embedding and referencing, making it a great choice for our application. By embedding related data within documents, we can optimize read performance, while referencing allows for flexibility and scalability when handling larger datasets.

User Authentication & Authorization

Secure user authentication is implemented using JWT (JSON Web Tokens). The authentication system includes user registration and login, token-based authentication, token management and updates and account deletion and token invalidation.

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Content Delivery & API Management

Content Delivery & API Management

The backend provides structured RESTful APIs to serve content dynamically. This includes story retrieval, quiz management, and progress tracking. All API responses follow a standardized JSON format to ensure consistency and ease of integration.

User Progress Tracking

Progress tracking is implemented using dedicated collections: Quiz Progress (quizProgress) and Story Progress (storyProgress). This enables personalized learning experiences, allowing users to track their achievements.

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Interactive Features & AI Integration

Interactive Features & AI Integration

To enhance engagement, the backend integrates AI-powered Q&A using Google Generative AI. Key features include chatbot assistance. This AI-driven approach improves accessibility and learning retention, ensuring that interactions are tailored to the educational content provided.

FastAPI Integration

This part of the application is built using FastAPI, a high-performance web framework for Python, ensuring efficient request handling and real-time responses. The FastAPI backend powers the model’s ability to process educator-uploaded data files, improving the accuracy of responses. It provides a robust API for configuring and interacting with the AI model, making it easily accessible for frontend integration.

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Authentication Endpoints Overview

User Registration

Endpoint for user registration. Accepts user details and creates a new account.

User Login

Endpoint for user login. Validates user credentials and generates authentication tokens.

User Logout

Endpoint for user logout. Invalidates user tokens and logs the user out of the system.

Delete Account

Endpoint for deleting user accounts. Allows users to permanently delete their account.

Password Reset

Endpoint for resetting user passwords. Sends a password reset link to the user's email.

Email Verification

Endpoint for email verification. Verifies the user's email address.

Edit User Profile

Endpoint for editing user information. Allows users to update their profile details.

Change Password

Endpoint for changing user passwords. Allows users to update their password.

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Model related Endpoints

Upload Data

This endpoint allows educators to upload course materials to the platform. It exclusively accepts PDF files, ensuring that the content is in a standardized format. All uploaded materials are stored securely in cloud storage.

Model Learning

This endpoint is used to extract text from uploaded PDF files. It takes in training data, sets up model settings, and starts the learning process. The extracted text is then sent to a Gemini session to provide answers based on the PDF content.

Answer Question

This endpoint answers user questions by querying the trained AI models. It takes the question and the age of the child to generate more suitable answers.

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Deployment & Infrastructure

1 Database Setup: MongoDB Atlas

The database for this application is deployed on MongoDB Atlas, a cloud-based NoSQL database service that provides high availability, scalability, and security. The database is hosted under the gp folder and is managed within Cluster 1. The primary database used in this project is named test, where all collections related are stored.

2 Version Control: GitHub

The project's codebase is stored and managed on GitHub, a widely-used version control platform. This helps us tracking changes, and maintaining a centralized repository for the application's source code. following a Git-based workflow, utilizing features like branches, pull requests, and code reviews to ensure code quality and seamless integration.

3 Content Storage: Cloudinary

The educational content PDFs are stored on Cloudinary, a cloud-based media management platform. The links to these files are stored in the database for easy retrieval and delivery.

4 Cloud Deployment: Vercel

To ensure seamless backend integration with the frontend team, the project is deployed on Vercel, a cloud platform that provides serverless hosting with automatic scaling. The deployment is divided into two separate services for better modularity and performance. The first deployment, named gpFirst, is dedicated to handling the machine learning model, which is responsible for learning processes and answering user questions. The second deployment, named gp, manages the core backend functionalities, including authentication, data uploading, OTP verification, and API routes.

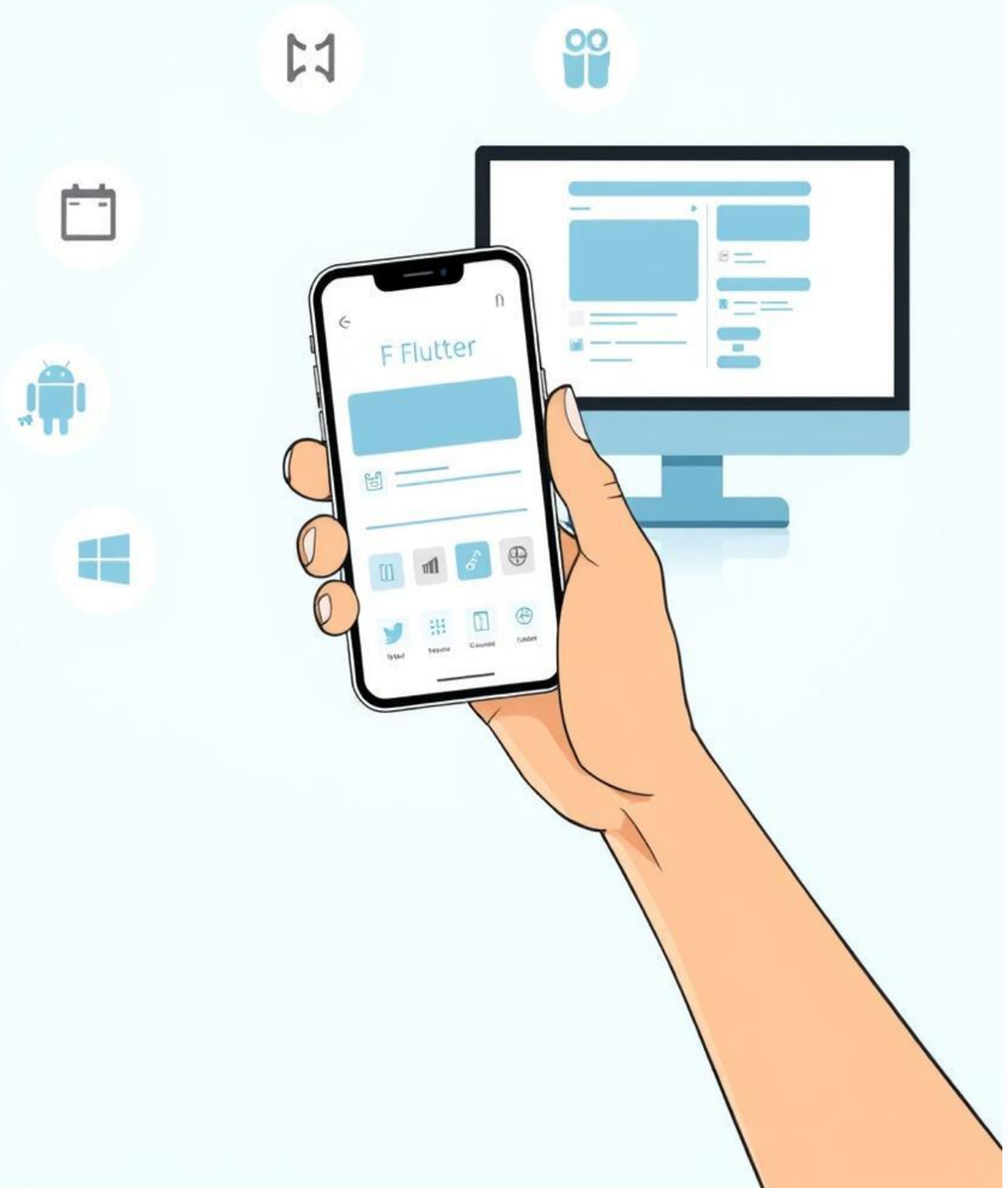
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KmtKids App: Frontend Overview

This part covers the frontend architecture of the KmtKids app. It highlights key technologies, structure, and features.

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Frontend Technologies

1 Flutter Framework

Open-source framework for cross-platform apps.

2 Dart Language

Used for building high-performance applications.

3 Key Libraries

Flutter/material, **SpeechToText, Flutter_tts** (text to speech), share_ preferences, FilePicker (upload file), http (to handle requests), Mimi, permission_handler (to ask user for permissions), and more...

4 Github

For version control and collaboration.

5 Figma

UI, UX design

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Welcome & Login Screens

Welcome Screen

Displays logo and checks token validity.

Navigates to Home or Login screen.

Login Screen

Verifies user credentials via API.

Stores tokens securely for future use.

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Registration & Confirmation

Register Screen

Collects user data and validates inputs.

API Submission

Sends validated data for registration.

Confirmation Screen

Confirms successful registration.

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Password Recovery Flow

1

Forgot Password

User enters email for reset request.

2

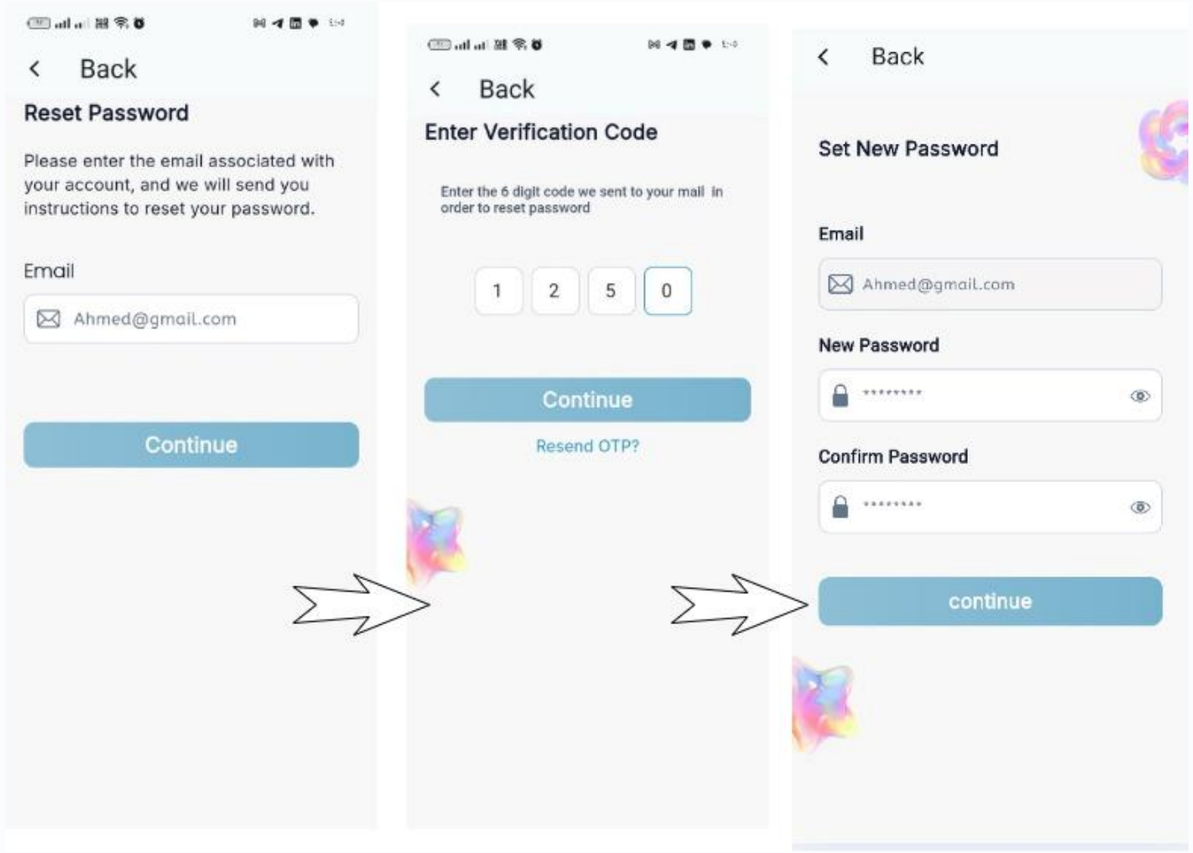
OTP Verification

User verifies identity with OTP.

3

Reset Password

User sets a new password.



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Interactive Chatbot

Text Messaging

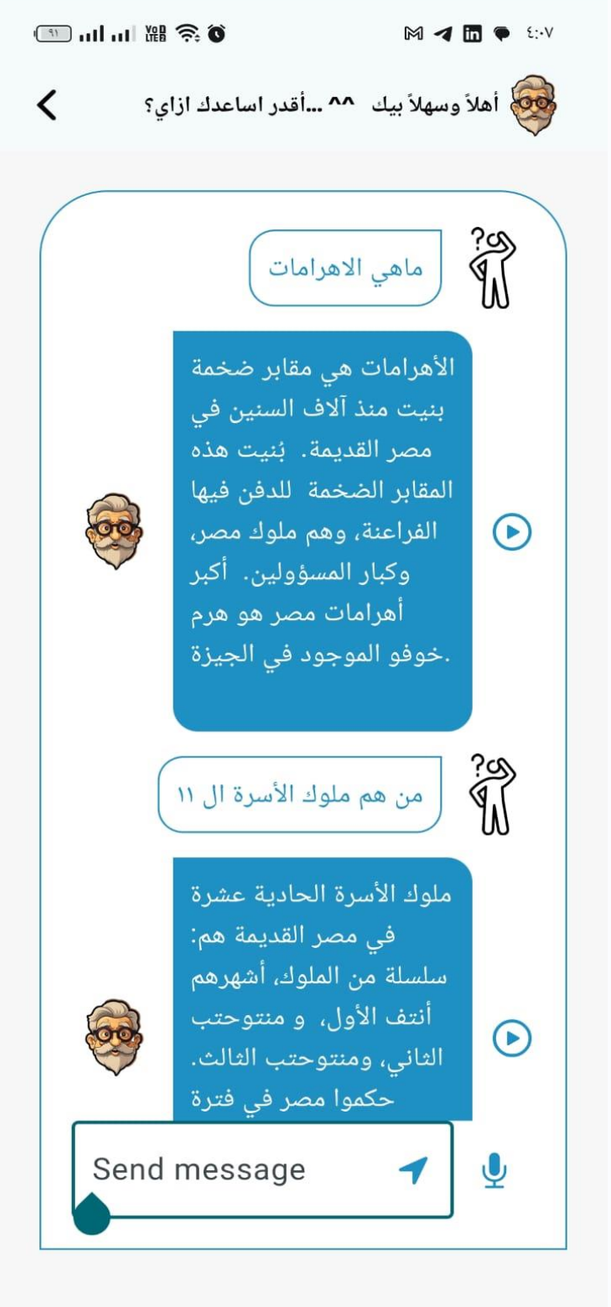
Users can type messages and ask questions

Voice Recognition

Converts speech to text using a library which used by user to ask questions

API Communication

Sends messages and displays responses.



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User Profile & Editing

Profile Screen

Displays user information and achievements.

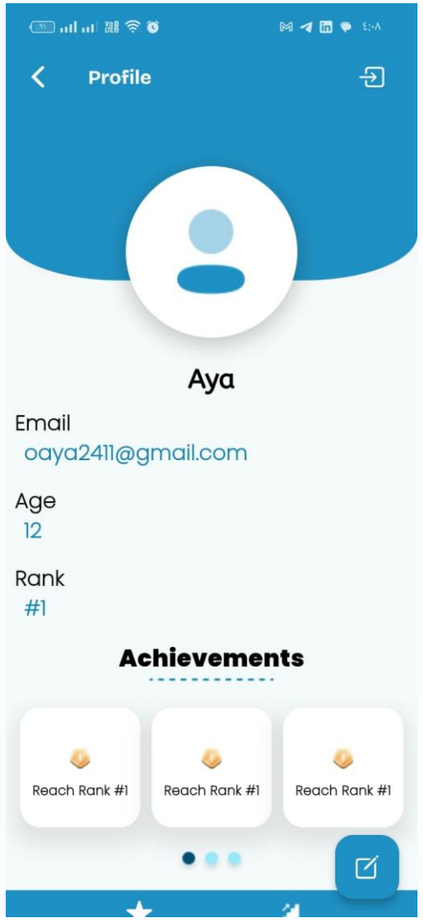
Edit Screen

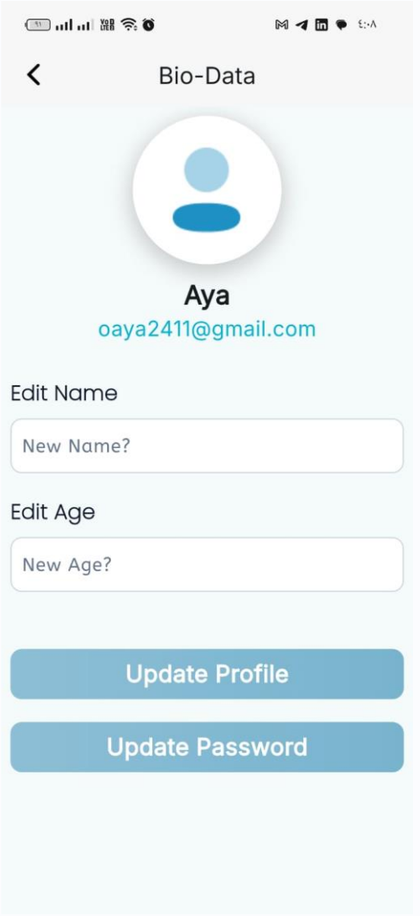
Allows users to modify their details.

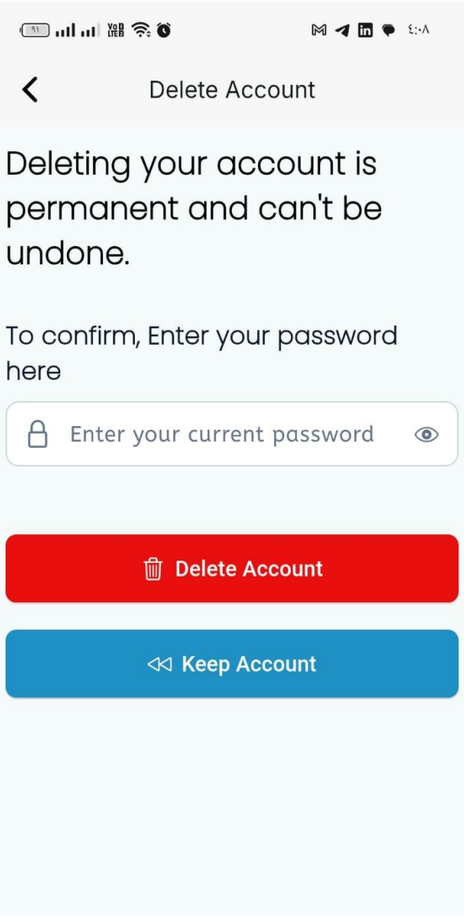
Account Options

Update password and delete account.

Logout from Account







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Home & Stories Screens

1

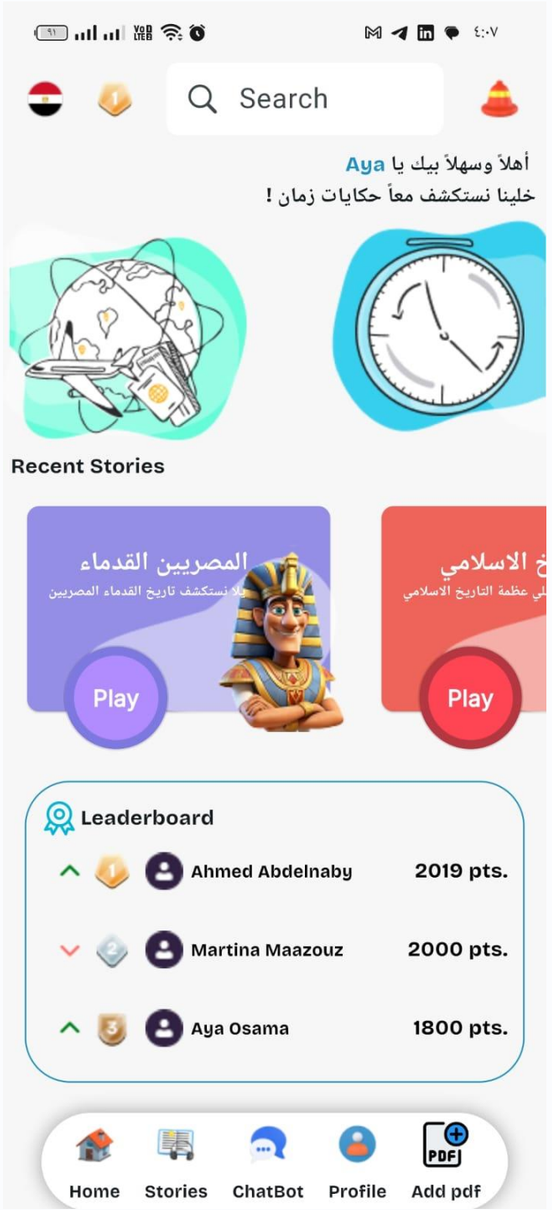
Home Screen

Displays real-time updates and user greetings.

2

Stories Screen

Presents historical stories interactively.



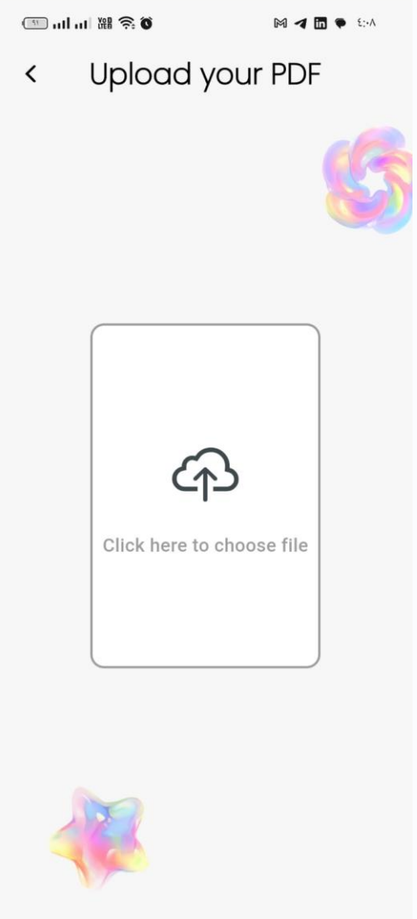


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File Upload & Enhancements

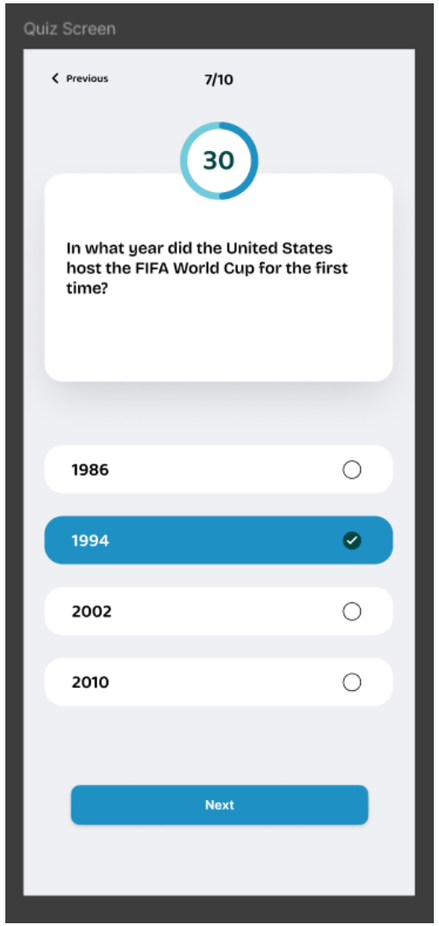
Upload File

Allows users to upload PDF documents.



Future Enhancements

Search, stories, quizzes, responsive design, and more.





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Future Work

We will improve personalized learning through AI by focusing on three areas:

1. **Dynamic Content:** AI will create custom learning content (quizzes, summaries, exercises) based on student needs and retrieve relevant materials.
2. **Adaptive Quizzes:** AI-powered quizzes will adjust to each student's level, providing detailed feedback and targeted support.
3. **Personalized AI Models:** We will develop AI models to understand each student's unique learning style, optimizing content delivery and strategies for continuous growth and satisfaction on all devices.



Dynamic Content Generation & Retrieval

Categories & Stories API

Optimized APIs for seamless content retrieval and organization. Each story category contains a predefined number of stories, ensuring structured content delivery.

AI-Driven Story Generation

Dynamic content creation based on predefined themes and topics, offering diverse and relevant educational material. Stories are dynamically created based on the theme.

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Automated Quizzes & Adaptive Scoring

1 AI-Powered Quiz Generation

Quizzes are dynamically created from story content, providing immediate comprehension assessment.

2 Varied Difficulty Levels

Questions accommodate different user levels, ensuring effective assessment of story comprehension.

3 Enhanced Validation Mechanisms

Improved quiz validation ensures content accuracy and engagement, fostering a reliable learning experience.

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Transformer-Based Models

Word Vectors

Numerical representations capture semantic meaning. Similar words have similar vector representations.

FastText Model

Breaks words into subwords and learns vectors for each. Excellent for morphologically rich languages like Arabic.

Mean Shift Algorithm

Finds dense regions. Groups sentences with similar semantic meanings into the same cluster.

Clustering helps generate stories by grouping related files, with file names from the backend.

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Personalized Learning & Custom AI Models

1

Custom AI Models

Develop custom AI models to enhance personalized learning experiences for each user.

2

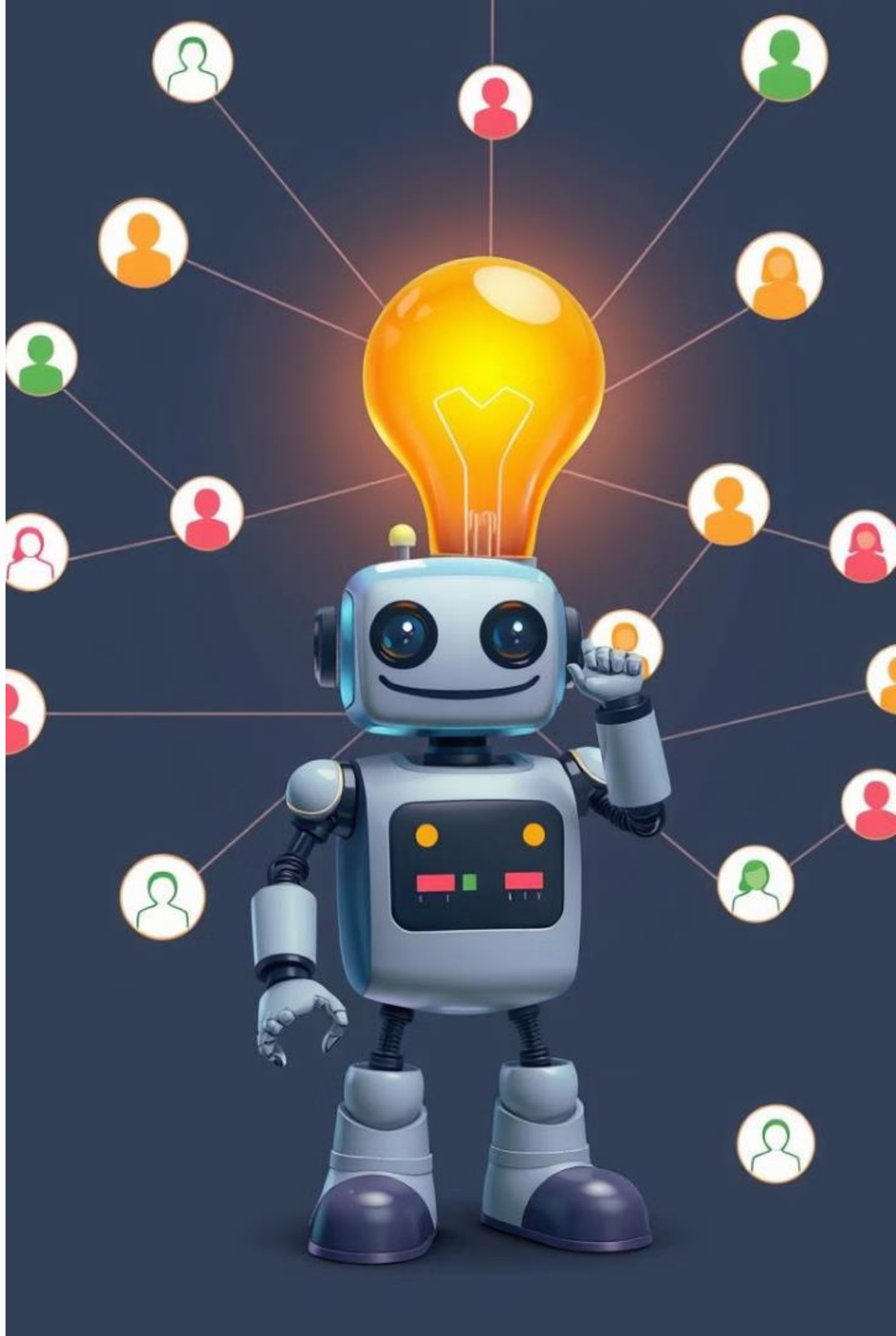
Transformers for Custom Model Development

Fine-tune a custom transformer-based model on domain-specific data for optimal performance.

3

RAG Integration

Incorporate external knowledge sources for tasks requiring factual accuracy and context-aware responses by integrating RAG.



Transformer-Based Models

- **Fine-Tuning LLaMA for Arabic Datasets:** Leverage LLaMA models, fine-tuning them on large-scale Arabic datasets to ensure accurate and contextually relevant responses in Arabic. This includes adapting the models to understand the nuances of different Arabic dialects and cultural contexts.
- **Evaluation of Fine-Tuned Transformers:** Rigorously evaluate the fine-tuned transformer models, focusing on key metrics such as relevance, retrieval efficiency, and coherence of generated content. Implement A/B testing to compare different fine-tuning strategies.
- **Domain Adaptation for Specific Applications:** Train transformer models on specific Arabic corpora, tailoring them for particular applications such as educational content generation, automated question answering, and personalized learning recommendations.
- **Efficiency Optimization through Quantization:** Implement quantization techniques to optimize the efficiency of transformer models, reducing their size and computational requirements, enabling deployment on devices with limited resources and improving response times.

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Retrieval-Augmented Generation (RAG)

1 Enhanced Accuracy in Arabic Content

By retrieving relevant passages from large-scale Arabic datasets, RAG significantly reduces inaccuracies and enhances the contextual relevance of generated content.

2 Real-Time Knowledge Updates

Instead of time-consuming model retraining, RAG enables dynamic content updates by simply modifying the indexed knowledge sources, ensuring responses remain current and accurate.

3 Custom Domain Integration

RAG facilitates the integration of custom domain knowledge sources, allowing transformer models to answer specific questions without extensive fine-tuning on niche datasets.

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How RAG Works

1

Embedding Generation

Convert text to vector representations for semantic meaning.

2

Vector Database Storage

Store embeddings in FAISS, Pinecone, or Weaviate.

3

Query Encoding

Convert user questions into embeddings.

4

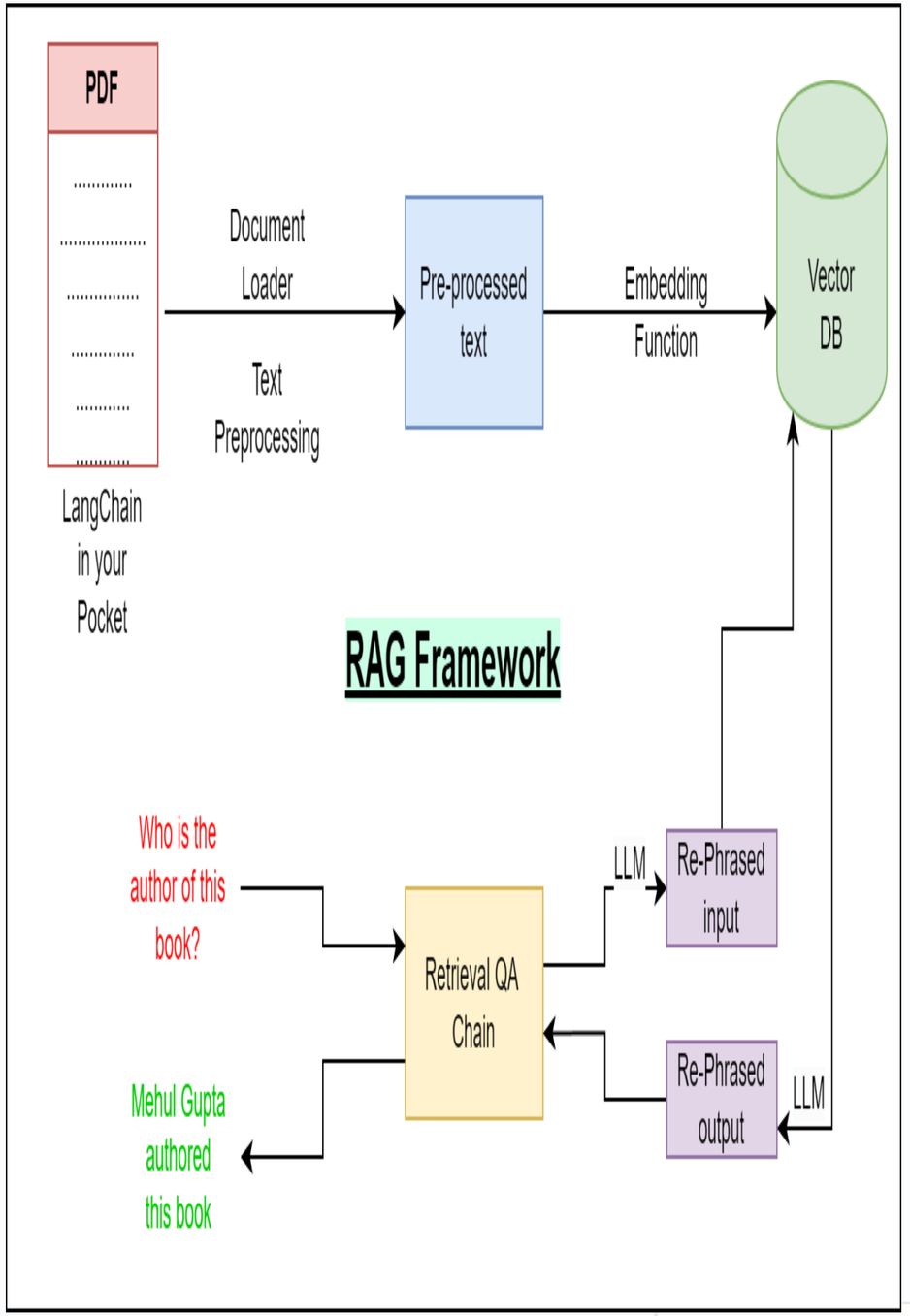
Retrieval Phase

Search the database for relevant documents.

5

Contextual Answer Generation

Generate answers using retrieved context.



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